

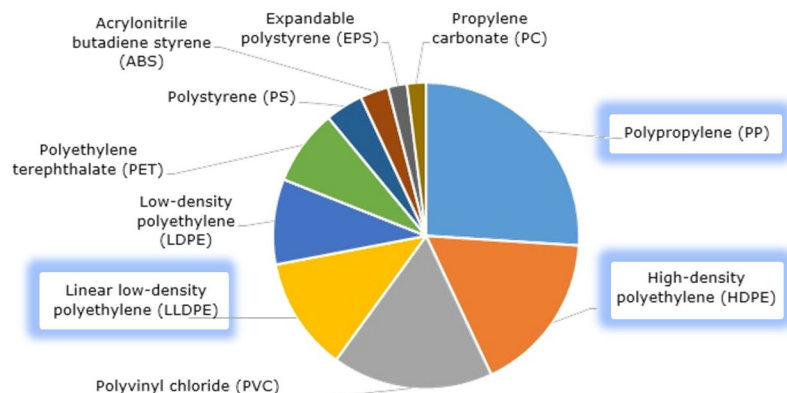
- Betrachtet man die Eigenschaften der anionischen Polymerisation als Vergleichsgrundlage, welche Vor-/Nachteile hat die radikalische Polymerisation?
- Was ist das allen RDRP-Methoden gemeinsame Konzept?
- Welche Kriterien müssen erfüllt sein, damit eine Polymerisation kontrolliert oder „lebend“ abläuft?
- Wie sieht das RAFT-Polymerisation-Gleichgewicht aus?
- Wie sieht die allgemeine Struktur eines RAFT-Agenten aus, und welche Z-Gruppen sind erforderlich, um die Polymerisation der einzelnen Hauptmonomerklassen (Methacrylate, Acrylate/Acrylamide, Styrole, Vinylester) zu steuern?
- Wie sieht das NMP-Gleichgewicht aus?

- Welche zwei Methoden gibt es, um eine NMP-Reaktion zu starten?
- Was ist der persistente Radikaleffekt (in ein oder zwei einfachen Sätzen)?
- Wie sieht das ATRP-Gleichgewicht aus?
- Was sind die Hauptkomponenten eines ATRP-Experiments?
- Wie sehen Initiatoren für ein effizientes ATRP von Methacrylaten aus?
- Mit welchen Methoden lässt sich die Konzentration von Übergangsmetallen bei der ATRP verringern?
- Warum kann die Reihenfolge der Sequenzen bei der RDRP-Synthese von Blockcopolymeren wichtig sein? Was passiert, wenn die Synthese in der „falschen“ Reihenfolge durchgeführt wird?

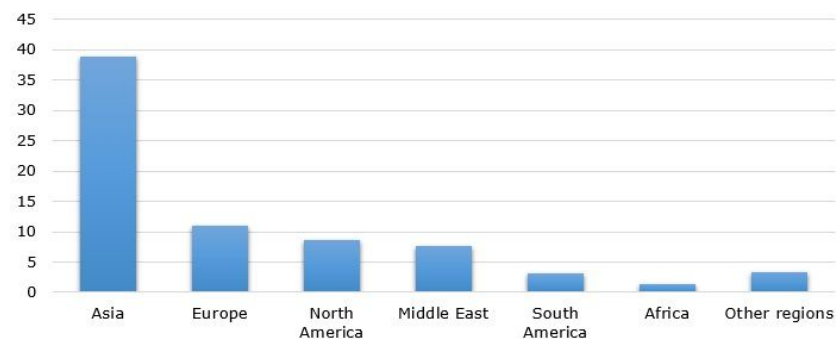
CHAPTER 4

Insertion polymerization; synthesis of polyolefins

Structure of global polymer demand by type (in volume), 2016



World's polypropylene production in 2016, by region (in million tons)



- Also known as „coordination polymerization“ or „catalytic polymerization“, and as a combination of two of three terms.

Note: „insertion polymerization“ is sometimes used to describe RDRP methods as well.

- Working for α -olefins
- Mechanism for close to 50 wt% of all polymers worldwide (all PP, most PE)

only possibility for PP!

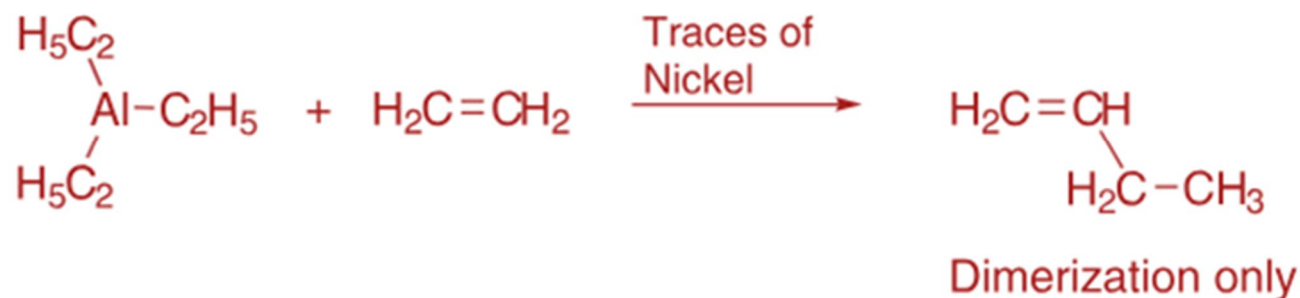
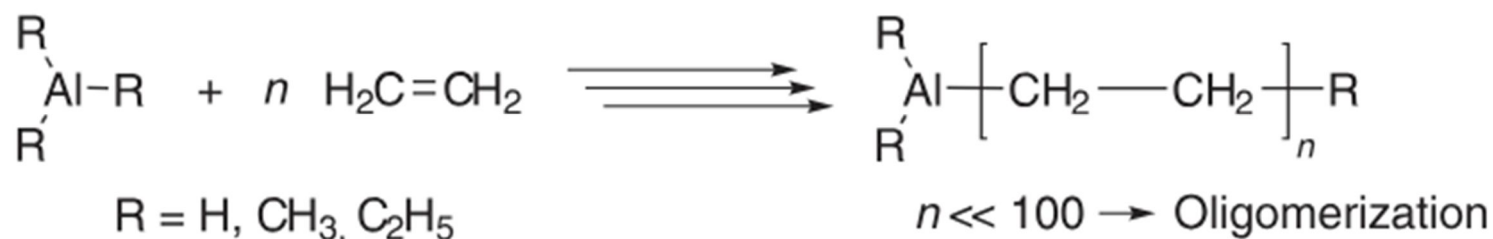


- Catalyst = transition metal, mostly Ti, Cr, and Zr, but also Fe, Ni, and Pd
- Co-catalyst required to provide the active metal center
- Control over tacticity ($\approx$ stereochemistry $\rightarrow$ crystallinity)

Insertion Polymerization

Historical Perspective

- **Karl Ziegler** (early 1950s; MPI Kofo, Mülheim an der Ruhr)

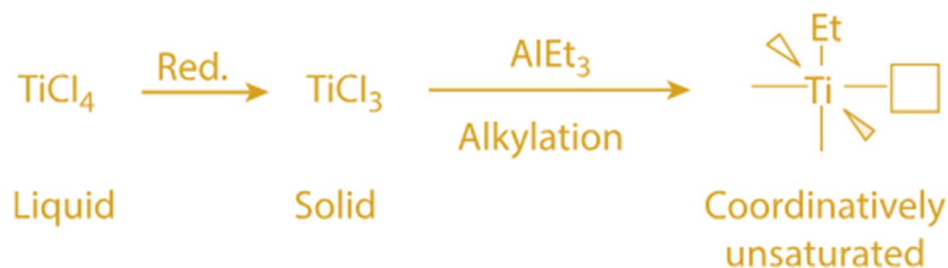


Insertion Polymerization

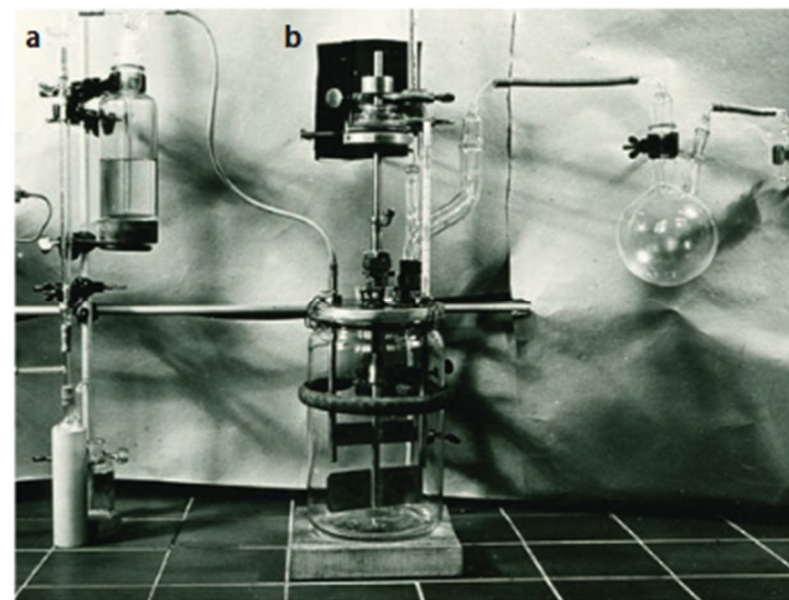
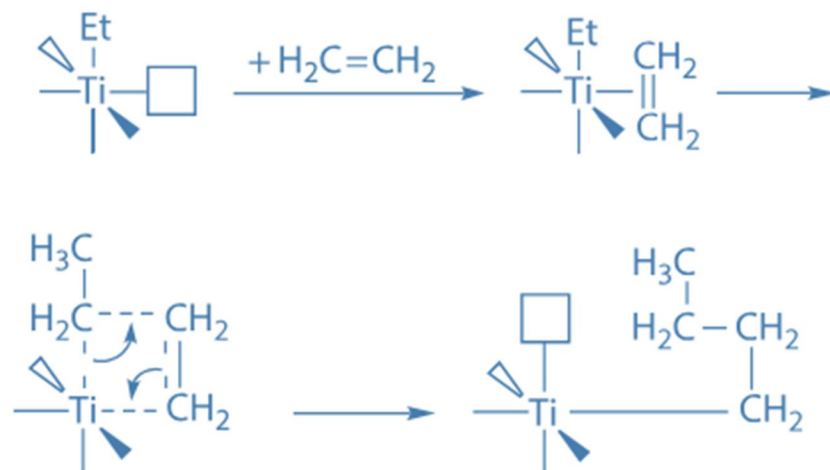
Historical Perspective

- **Karl Ziegler** (1953*; MPI Kofo, Mülheim an der Ruhr)
“Muelheim polyethene low-pressure process”

Initiation



Propagation



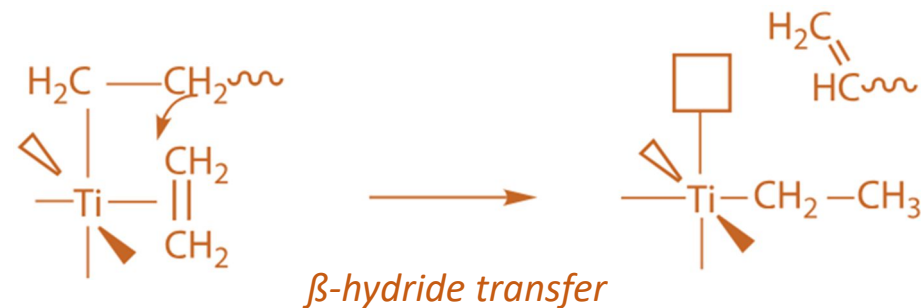
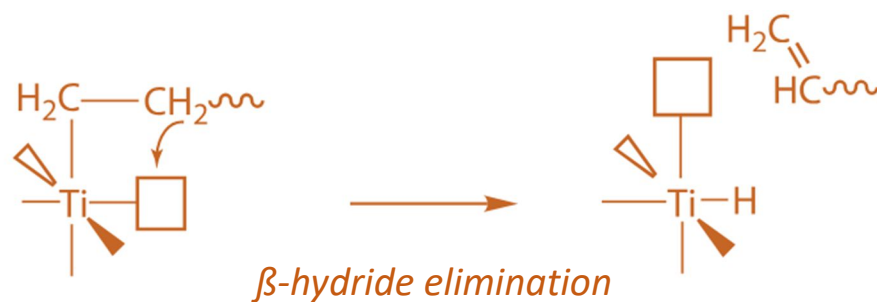
*patent filed on 17.11.1953; first presentation on 22.07.1955
Angew. Chem. Int. Ed. **2003**, 42, 5000

Insertion Polymerization

Historical Perspective

- **Karl Ziegler** (1953; MPI Kofo, Mülheim an der Ruhr)
“Muelheim polyethene low-pressure process”

Chain transfer



Termination

Due to contamination, e.g., by water

Control of molar mass

$$\text{Kinetic chain length } \nu = \frac{\text{Olefin insertion rate}}{\text{Rate of termination and transfer}}$$

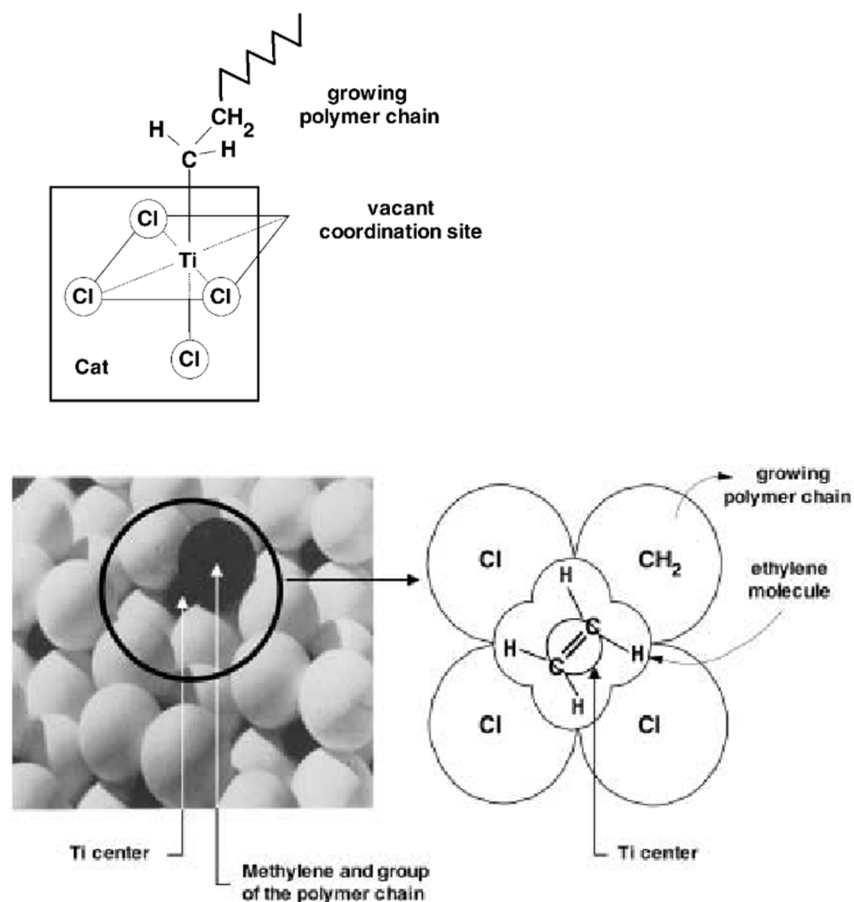
Regulated through temperature, ethylene pressure, addition of hydrogen.

Insertion Polymerization

Historical Perspective

- Karl Ziegler (1953; MPI Kofo, Mülheim an der Ruhr)
“Muelheim polyethene low-pressure process”

Catalyst structure



Reviews

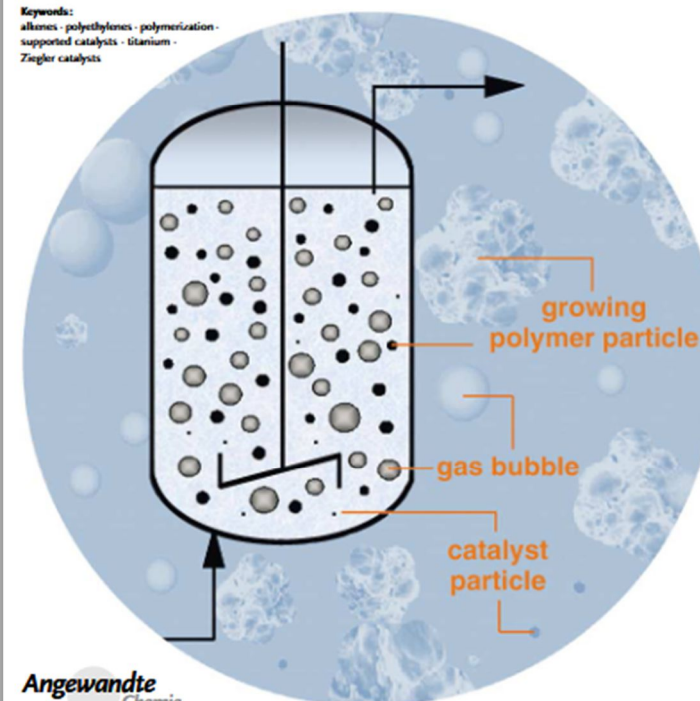
L. L. Böhm

Fifty Years of Ziegler Catalysts

The Ethylene Polymerization with Ziegler Catalysts: Fifty Years after the Discovery

Ludwig L. Böhm*

Keywords:
alkenes · polyethylenes · polymerization ·
supported catalysts · titanium ·
Ziegler catalysts



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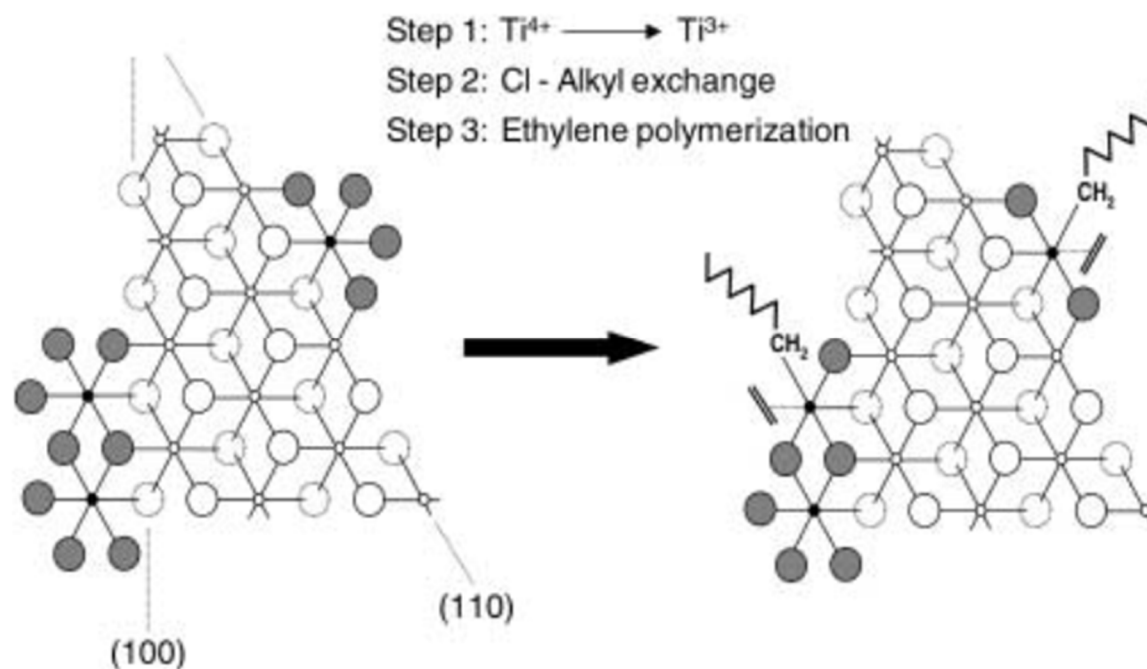
Insertion Polymerization

Historical Perspective

- Karl Ziegler (1953; MPI Kofo, Mülheim an der Ruhr)
“Muelheim polyethene low-pressure process”

Supported catalysts

1968 (Montedison S.p.A, Italy) – TiCl_4 adsorbed onto MgCl_2 , then reduced and activated with alkyl aluminium



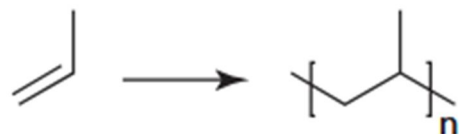
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Insertion Polymerization

Historical Perspective

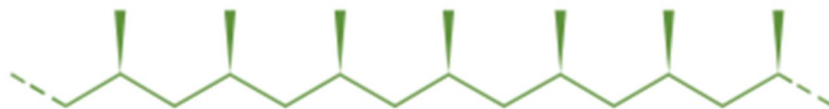
- **Giulio Natta** (1954; Politecnico di Milano)

Extension to propylene and other α -olefins... leading to polyolefins soluble in the reaction medium (*atactic*)



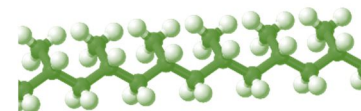
polypropylene

„isotattico“



Isotactic

$T_g = 0\text{ }^{\circ}\text{C}$; $T_m = 184\text{ }^{\circ}\text{C}$
helix-forming

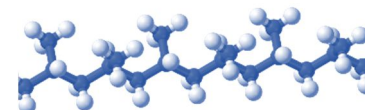


„sindiotattico“

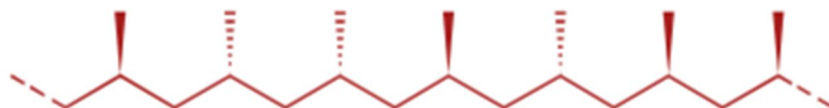


Syndiotactic

$T_g = -8\text{ }^{\circ}\text{C}$; $T_m = 160\text{ }^{\circ}\text{C}$



„atattico“



Atactic

$T_g = -10\text{ }^{\circ}\text{C}$; $T_m = -$

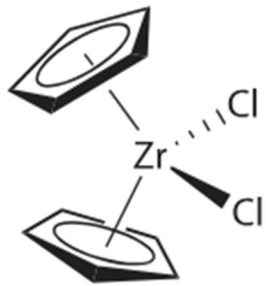


Rosita Beati (Natta's wife) $\rightarrow$ Τακτικός = taktikos: "relating to arrangement or order"

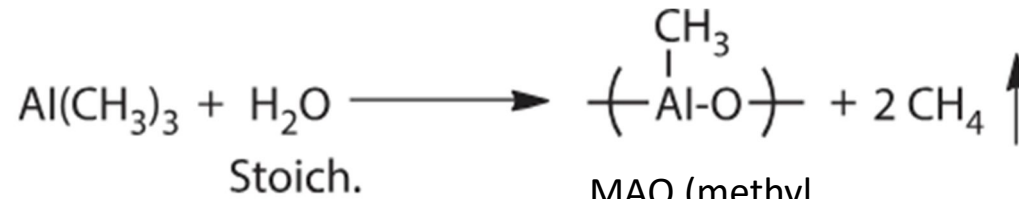
Insertion Polymerization

Homogeneous Catalysts

= Metallocenes and related (start in 1995; Brintzinger)

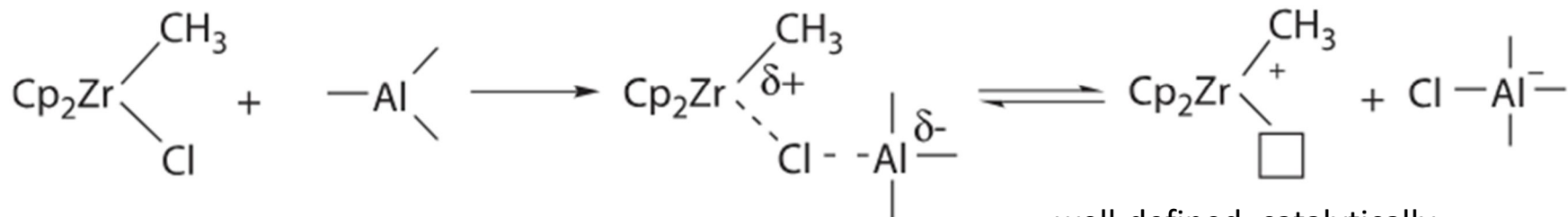
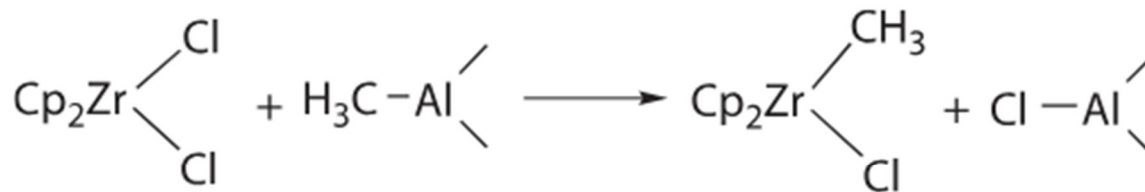


catalyst



MAO (methyl
aluminoxane)

co-catalyst



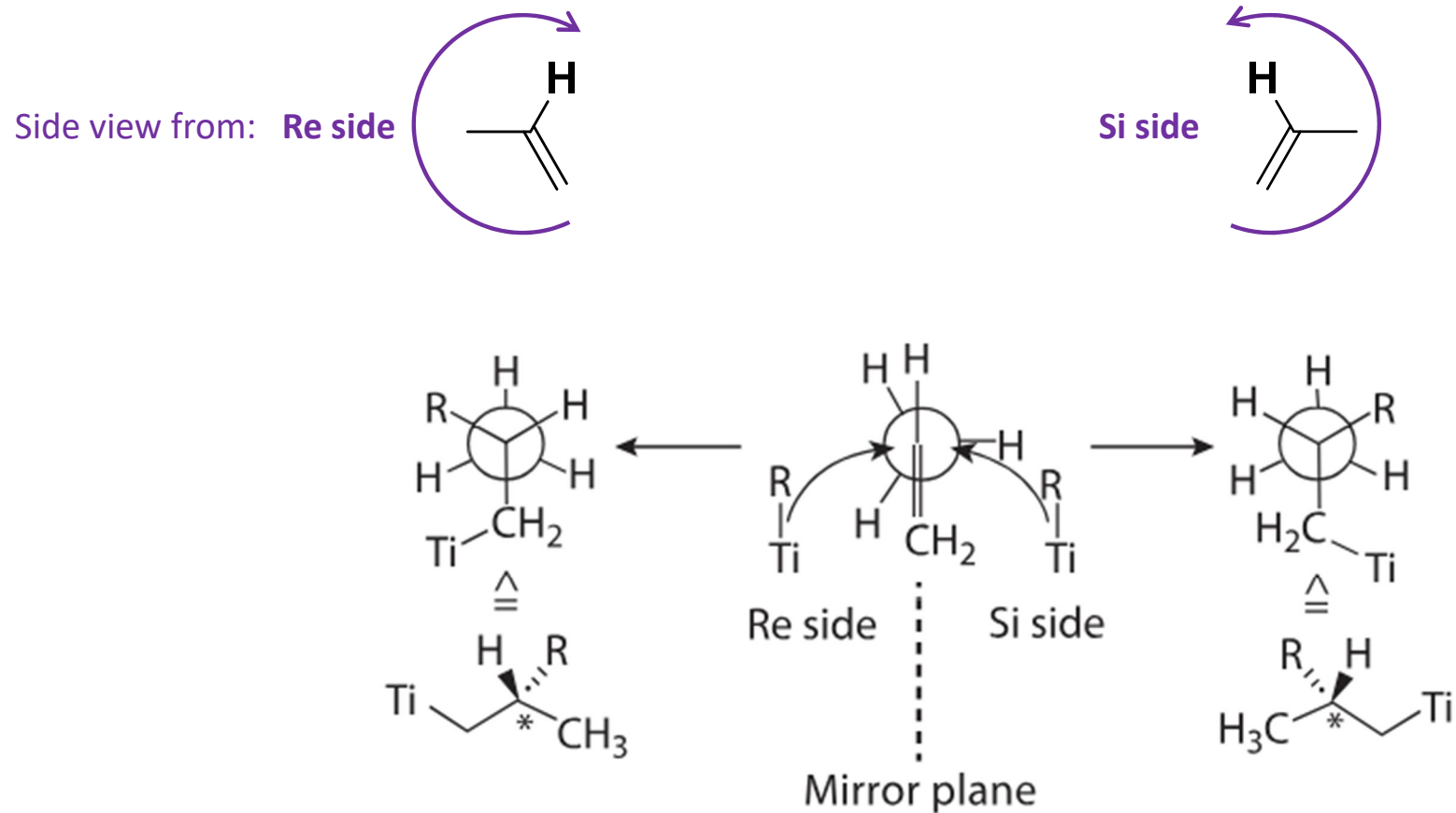
well-defined, catalytically
active component
→ **single-site catalyst**

Insertion Polymerization

Homogeneous Catalysts

= Metallocenes and related (start in 1995; Brintzinger)

→ Offer straightforward pathway to stereochemical control

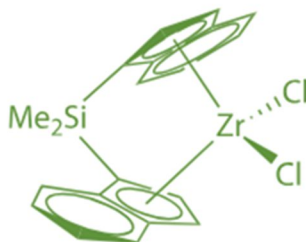


Insertion Polymerization

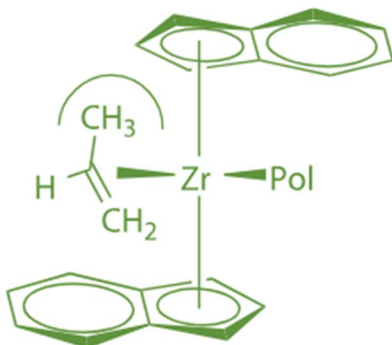
Homogeneous Catalysts

= Metallocenes and related

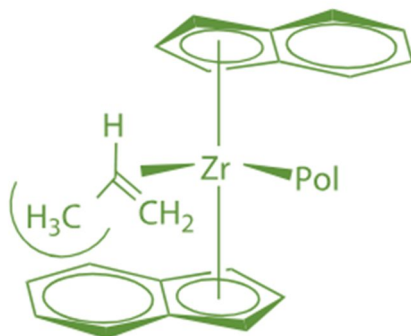
Metallocenes for isotactic PP



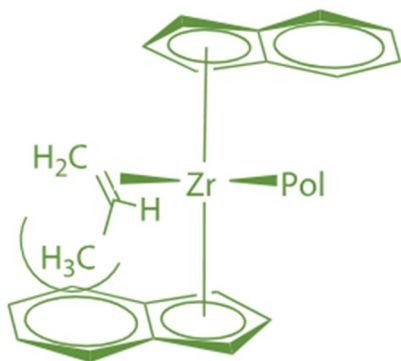
Options for the approach of propylene



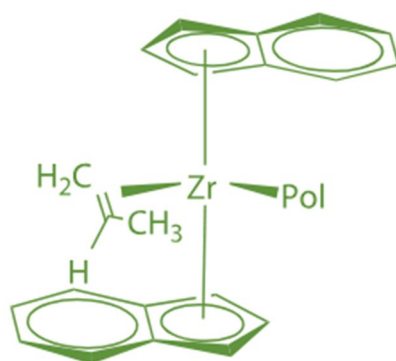
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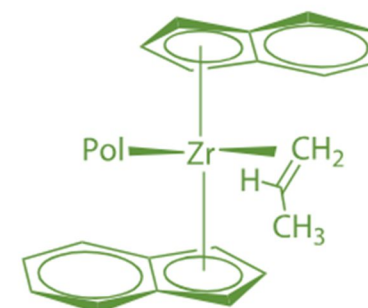
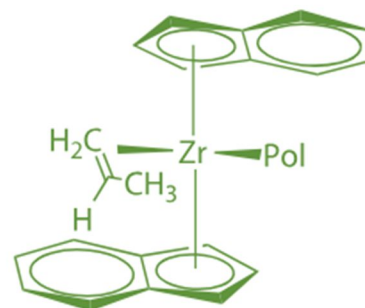
II



III



IV



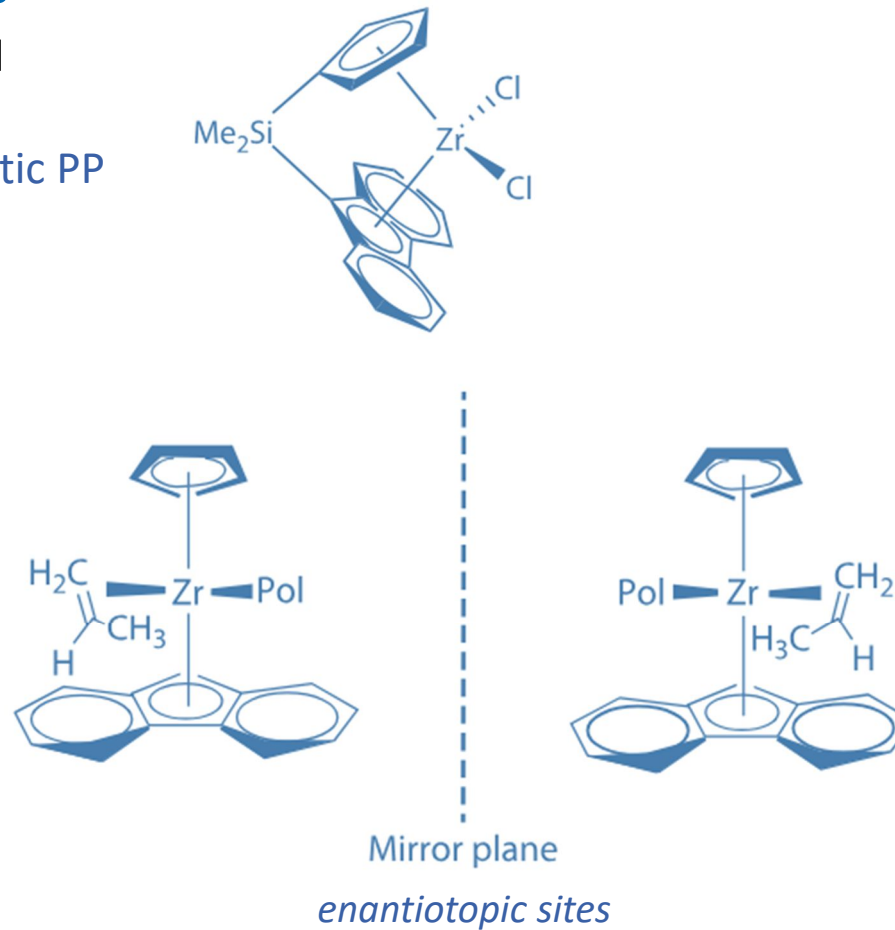
homotopic sites

Insertion Polymerization

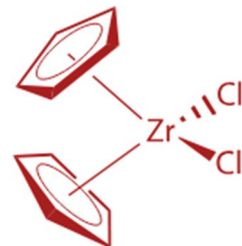
Homogeneous Catalysts

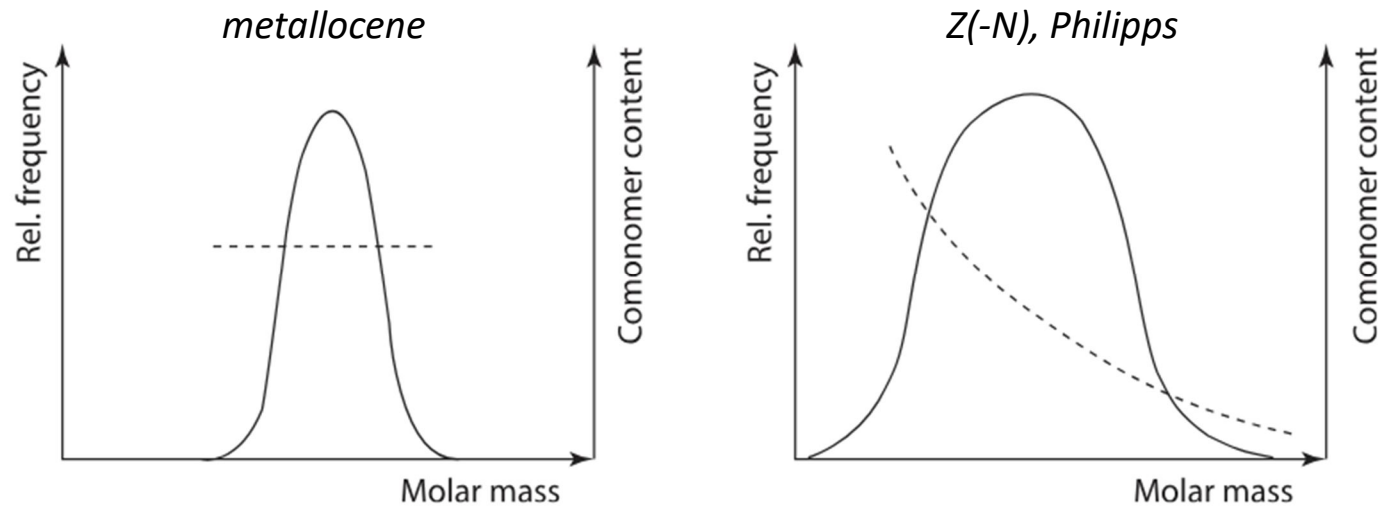
= Metallocenes and related

Metallocenes for syndiotactic PP



Metallocenes for atactic PP





Drawbacks of metallocenes

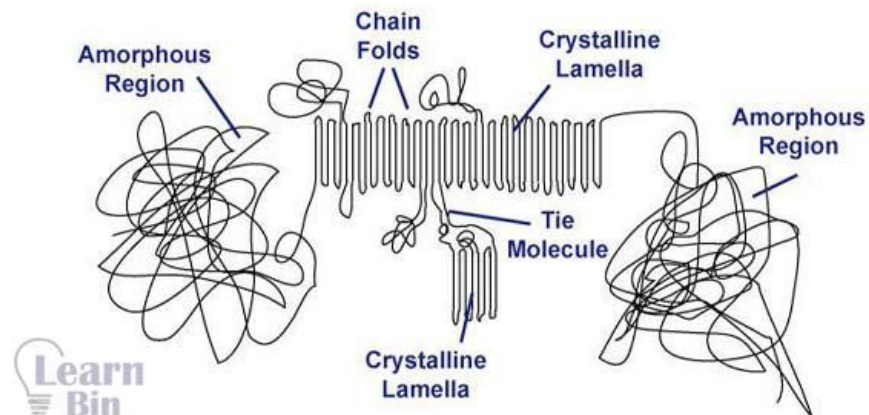
- price
- past patent issues limiting broad technical use

Advantages of metallocenes

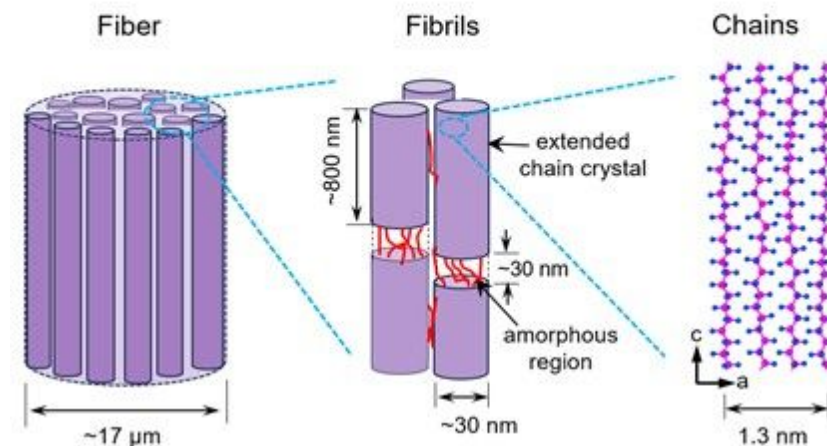
- higher incorporation of comonomers and more homogeneous comonomer distribution (*e.g.*, for ethylene with 1-hexene or 1-octene)
- lower dispersity ($\bar{D} \approx 2$ vs $\bar{D} = 5-10$)
- less oligomers
- PP in all stereo-structural variations with high purity
- block copolymers by “chain shuttling”

UHMWPE

LAMELLA STRUCTURE UHMWPE



Multiscale structure of a ultra-high molecular weight polyethylene fiber



Cycloolefin Copolymers

